October 30, 1967
SUPERSEDING
Int. Fed. Spec. 000-W-00645a (GSA-FSS)
August 25, 1964

INTERIM FEDERAL SPECIFICATION

WRENCH; OPEN END BOX (AND BOX COMBINATION, CROWFOOT, RATCHET, AND RATCHET CROWFOOT)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Standardization Division, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers non-distortion types of wrenches for use on tube fittings, stuffing gland nuts, and applications where difficulties arise in the use of standard design wrenches.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the titles of the specifications, or which are commercially available, but include only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types. Wrenches shall be furnished in the following types, as specified (see 6.1):

Type I - Open end box and box combination.
Type II - Open end box crowfoot.
Type III - Open end box ratchet.
Type IV - Open end box ratchet crowfoot.
Type V - Open end box, double end.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specification:

QQ-A-225/5 - Aluminum Alloy Bar, Rod and Wire (Rolled or drawn) 2017.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 151 - Metals, Test Methods.

FSC 5120

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-H-15424 - Hand Tools, Packaging of.
MIL-F-18866 - Fittings, Flared Tube, 37 degree, steel.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or requests for proposal shall apply.

United States of America Standard Institute (USASI):
(USASI) B18.2-1960-Square and Hexagon Bolts and Nuts.

(Application for copies should be addressed to the U. S. of American Standards Institute, (USASI), 10 East 40th Street, New York, New York 10016.)

3. REQUIREMENTS

3.1 Material. The wrenches shall be steel, the composition and heat treatment of which shall be such as to provide wrenches conforming to the requirements specified herein.

3.2 Design. Wrenches shall be similar to the figure to which reference is made with openings of the proper size to slide over the maximum size regular hex heads referenced in USASI B18.2-1960, as applicable.

3.3 Test loads. Wrenches shall withstand the test loads specified in the applicable tables without injury or permanent deformation which may affect their durability or serviceability.

3.4 Performance tests. The wrenches shall meet the tube fitting distortion and aluminum mandrel tests as specified in the applicable tables when performed in accordance with 4.6.2.2 and 4.6.2.3. In addition, the box end of type I wrenches shall resist, without injury, the steel mandrel test of 4.6.2.1.

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3.5 Hardness. Wrenches shall be hardened to Rockwell C 40 minimum (see 4.6.1).

3.6 Finish.

3.6.1 Surface. All surfaces shall be substantially free from pits, nodules, cracks, and other defects. Areas ground, buffed, or finished by an equivalent method shall have a bright finish with a maximum surface roughness of 32 microinches when chromium plate is specified (see 6.1), or 64 microinches maximum when other than chromium is furnished. Chromium plated wrenches shall be ground or buffed for approximately 180° of the head periphery. Surfaces not ground or buffed shall have a surface conforming to good commercial practice and the surface roughness shall not exceed 125 microinches.

3.6.2 Coating. The coatings shall be adherent, smooth, continuous, and free from uncoated areas, pits, blisters, nodules, and other defects which would interfere with their protective value and serviceability. Unless otherwise specified (see 6.1) wrenches shall have one or a combination of the coatings listed herein; if chromium is specified but restricted by Government order, one of the other listed coatings will be acceptable. The movable members of ratchets need not be plated.

3.6.2.1 Bright chromium plate. The finish shall be electrodeposited metals consisting of nickel, followed by chromium, the minimum thickness being 0.0002 inch for nickel and 0.00001 inch for chromium.

3.6.2.2 Bright nickel plate. The finish shall be an electrodeposited coating of nickel and shall be not less than 0.0002 inch.

3.6.2.3 Cadmium plate. The finish shall be an electrodeposited coating of cadmium not less than 0.0003 inch thick, and shall be subjected to a chemical or electrochemical chromating.

3.6.2.4 Zinc plate. The finish shall be an electrodeposited coating of zinc not less than 0.0003 inch thick, and shall be chromated.

3.6.2.5 Hot phosphating. The finish shall consist of a heavy chemically produced phosphate coating followed with a coating of preservative.

3.6.2.6 Oxide coating. The finish shall consist of a heavy chemically produced oxide coating followed with a coating of preservative.

3.7 Sets. When specified (see 6.1) wrenches shall be furnished in sets of the type and size specified (see 6.1). Sets shall be furnished in fabric wraps or rolls, or in boxes, as specified (see 6.1). The wraps or rolls shall contain compartments for each tool in the set and shall be furnished with durable ties. The boxes for sets shall be made of sheet metal, shall be substantial, durable, and of rigid construction, and shall be provided with either a catch or hasp and staple for retaining a hinged cover in a closed position. The interior and exterior surfaces of each box shall have a durable coating of paint or enamel.

3.8 Marking. Each wrench head shall be marked in a plain and permanent manner with the respective wrench opening as shown in the first column of the applicable table. In addition, each wrench shall be marked in a plain and permanent manner with the manufacturer's name or trademark of such known character that the source of manufacture may be determined.

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3.9 Type I open end box and box combination. The type I shall be a double end wrench of the same nominal size on each end. One end shall contain a cut out. The wrench shall be similar to figure 1 and conform to table I.

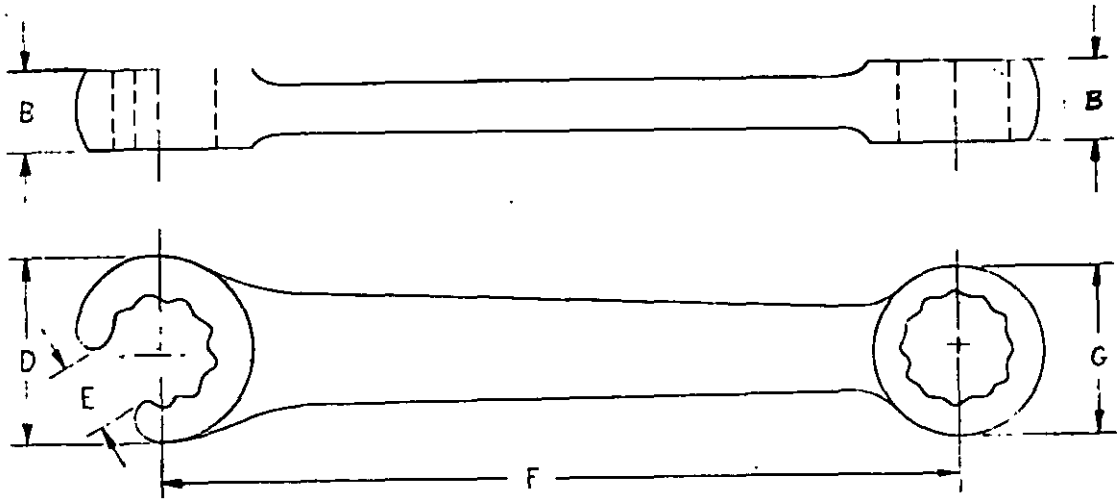


FIGURE 1. Type I wrenches.

TABLE I. Types I and II wrenches

Nominal size	A		H	B	D	E	F	G	Test loads inch-pounds (Min.)		
(inch)	Min. (inch)	Max. (inch)	Min. (inch)	Max. (inch)	Max. (inch)	Min. (inch)	± 1/2 (inch)	Max. (inch)	Tube fitting test	Steel mandrel test (box end)	Aluminum mandrel test
3/8	.378	.385	.505	.312	.859	.193	5-1/4	.625	240	550	180
7/16	.378	.385	.505	.343	.875	.255	5-1/2	.718	250	650	260
1/2	.378	.385	.505	.437	.937	.317	5-7/8	.812	*	1020	320
9/16	.378	.385	.505	.440	1.031	.317	6-1/2	.906	330	1500	380
5/8	.503	.511	.670	.470	1.125	.380	7	1.000	360	2000	450
11/16	.503	.511	.670	.500	1.203	.380	7-5/8	1.093	540	2300	560
3/4	.503	.511	.670	.562	1.281	.380	8-1/2	1.165	*	2700	720
7/8	.503	.511	.670	.625	1.453	.505	10-3/8	1.343	600	3300	1200
15/16	.503	.511	.670	.625	1.531	.505	11-1/4	1.437	*	4100	1320
1	.503	.511	.670	.690	1.593	.630	11-5/8	1.531	720	4900	1400
1-1/8	.503	.511	.670	.781	1.781	.630	12-3/4	1.718	*	5900	1450

* No test required.

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3.10 Type II open end box crowfoot. The type II wrench shall conform to the requirements for the type I wrench except that one end shall be provided with a square hole. The wrench shall be similar to figure 2 and conform to table I, as applicable.

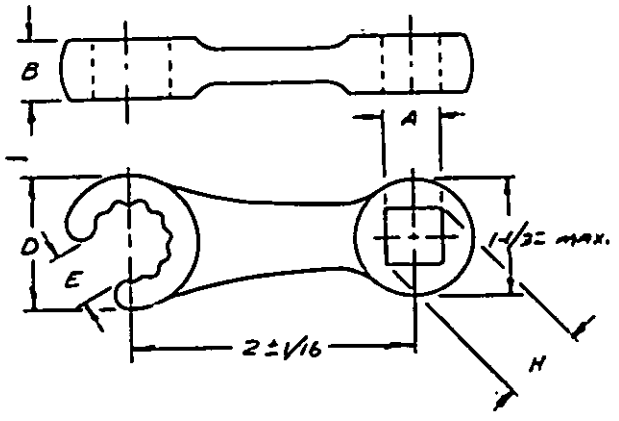


FIGURE 2. Type II wrench.

3.11 Type III open end box ratchet wrench. The type III wrench shall consist of a steel handle and ratchet mechanism. The ratchet mechanism shall consist of a series of spring loaded hardened steel rollers. The steel rollers shall be retained in a floating manner so as to create a cam action ratcheting effect for removing or replacing hexagon tube fittings. A hole at least 1/4 inch in diameter shall be provided at the handle end. The wrench shall be similar to figure 3 and conform to table II, as applicable.

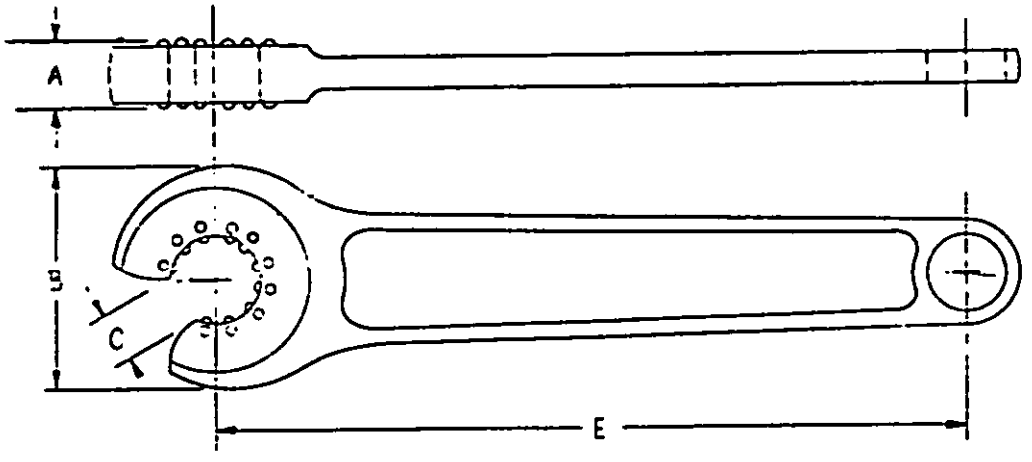


FIGURE 3. Type III wrench.

3.12 Type IV open end box, ratchet crowfoot. The wrench shall conform to 3.11 except that the end opposite the ratchet head shall be provided with a square hole. The wrench shall be similar to figure 4 and conform to table II.

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TABLE II. Types III and IV wrenches

Nominal size	A	B	C	E	F		G	Test loads inch-pounds (Min.)	
								Tube fitting test	Aluminum mandrel test
(inch)	Max. (inch)	Max. (inch)	Min. (inch)	± .25 (inch)	Min. (inch)	Max. (inch)	Min. (inch)		
3/8	.480	1.220	.255	4.75	.378	.385	.505	240	180
7/16	.480	1.280	.255	4.75	.378	.385	.505	250	260
1/2	.480	1.400	.317	4.75	.378	.385	.505	*	320
9/16	.480	1.560	.317	5.40	.378	.385	.505	330	380
5/8	.600	1.760	.380	6.00	.503	.511	.670	360	450
11/16	.600	1.850	.380	6.60	.503	.511	.670	540	560
3/4	.600	1.930	.505	7.20	.503	.511	.670	*	720
7/8	.600	2.250	.630	8.40	.503	.511	.670	600	1200
1	.600	2.450	.630	9.60	.503	.511	.670	720	1400

*No test required.

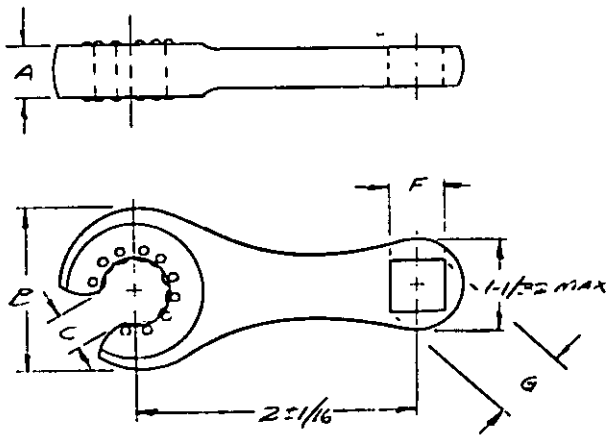


FIGURE 4. Type IV wrench.

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3.13 Type V open end box, double end. Each end of the wrench shall contain a cut out. The wrench shall be similar to figure 5 and conform to table III.

TABLE III. Type V wrench

Nominal size (inch)		B Max. (inch)		D Max. (inch)		E Min. (inch)		F ± 1/2 inch	Test loads inch-pound (Min.) Tube fitting test		Aluminum mandrel test	
Small end	Large end	Small end	Large end	Small end	Large end	Small end	Large end		Small end	Large end	Small end	Large end
3/8	7/16	.312	.343	.859	.875	.193	.255	4-7/8	240	250	180	260
1/2	9/16	.437	.440	.937	1.031	.317	.317	5-5/8	*	330	320	380
5/8	11/16	.470	.500	1.125	1.203	.380	.380	6-1/4	360	540	450	560
3/4	7/8	.562	.625	1.281	1.453	.380	.505	7	*	600	720	1200

*No test required.

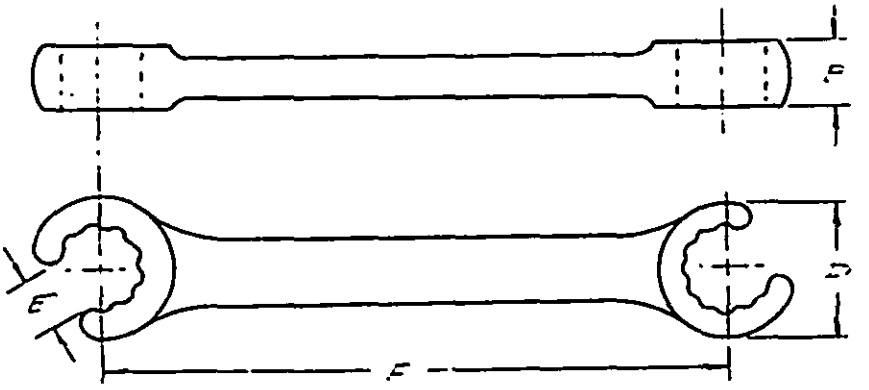


FIGURE 5. Type V wrench.

3.14 Scope of illustrations. The illustrations shown herein are descriptive and not restrictive, and except for dimensions, are not intended to preclude the purchase of wrenches which are otherwise in accordance with the requirements of this specification.

3.15 Workmanship. Workmanship shall be of the highest grade throughout and equal in every respect to good commercial practice. The wrenches shall be free from rust, burrs, fins, and any other imperfections which may impair their serviceability or durability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection lot. All wrenches and containers of the same type, presented at one time shall be considered a lot for purposes of acceptance, inspection, and tests.

4.3 Sampling for lot acceptance.

4.3.1 Sampling for visual and dimensional inspection. A random sample of wrenches shall be taken from each inspection lot of material offered for inspection of visual and dimensional characteristics in accordance with MIL-STD-105 at Inspection Level II and Acceptable Quality Level (AQL) shall be 1.5 percent defective.

4.3.2 Sampling for lot acceptance tests. A random sample of wrenches shall be taken from each inspection lot in accordance with MIL-STD-105 at inspection level S-4 and shall be subjected to the tests specified in 4.5, and the AQL shall be 1.5 percent defective.

4.4 Inspection

4.4.1 Visual and dimensional inspection. Each of the sample items selected in accordance with 4.3.1 shall be visually and dimensionally inspected to verify compliance with the requirements of this specification.

4.5 Lot acceptance tests. Each of the sample items taken in accordance with 4.3.2 shall be tested in accordance with 4.6 to verify compliance with the requirements of this specification.

4.6 Test procedures.

4.6.1 Hardness test. The wrench hardness requirements shall be tested in accordance with Method 243 of Fed. Std. 151 to determine compliance with 3.5.

4.6.2 Load test. The test load shall be applied as far from the wrench opening as practical, and sustained for 10 seconds. Care should be exercised to maintain accuracy in applying the load and measuring the effective lever arm. The effective lever arm, when the load is vertical, is the shortest distance between two vertical lines passing, respectively, through the point of load application and the center of the mandrel fitting the wrench opening. The bending moment shall be the product of the effective lever arm, in inches, and the load, in pounds. There shall be no permanent visible distortion of the wrench after completion of the tests.

4.6.2.1 Steel mandrel load test. The box end of type I wrenches shall be tested for conformance with 3.3 by applying the load specified in table I for steel mandrels. The mandrels shall be hardened to Rockwell C 55 minimum and conform to the dimensional requirements of USASI B18.2-1960 for the "across the flats" dimension. The length of the mandrels shall be sufficient to fully engage the wrenching surfaces.

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4.6.2.2 Tube fitting distortion test. A threaded steel tube nut fitting conforming to MIL-F-18866B shall be inspected with an NF-3 go, no go thread plug gage. The go gage shall be screwed into the nut until it bottoms (a light oil may be used) and backed off one revolution. The wrench to be tested shall be placed on the nut and located at the approximate center (see fig. 6). The nut, with the wrench and gage in position, shall be held by a socket wrench of the same size as the test wrench. The socket shall be secured to a torque analyzer. The test wrench shall be rotated until 1-1/2 the required test load is reached. This shall be done three times, each time rotating the wrench to the next gripping position. While constantly rotating the thread gage by hand, clockwise and counterclockwise, slowly increase the torque on the test wrench to the point where the gage cannot be turned by hand. Record this value. Repeat this procedure six times and average the lowest three values. This figure should be no lower than the test load specified in table I or II.

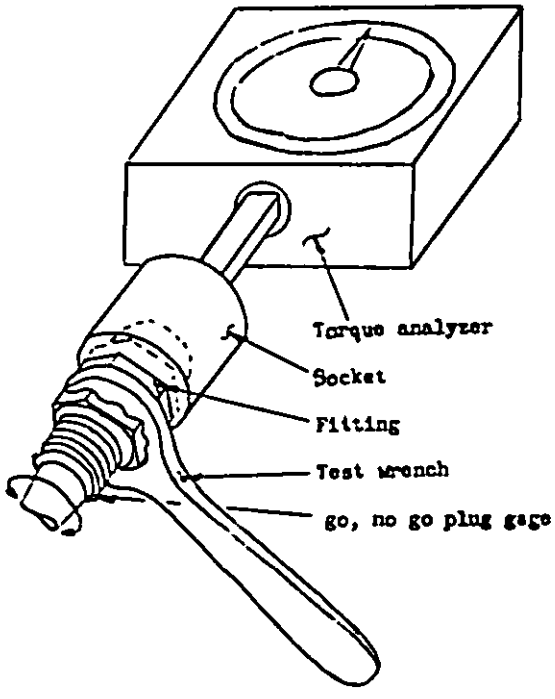


FIGURE 6. Test set-up for tube fitting distortion test.

4.6.2.3 Aluminum mandrel test. All wrench openings of the sample selected shall be tested on aluminum mandrels to determine their capabilities. The mandrels shall be made from aluminum conforming to QQ-A-225/5 in condition T-4 for all sizes under 1/2 inch, and condition T-451 for other sizes. The mandrels shall conform to figure 7. The wrench shall be placed on the mandrel and located centrally to the .140 inch long portion of the hexagon (see fig. 6). Care shall be exercised to maintain the face of the wrench parallel to the face of the mandrel during the test. Force shall be applied to the wrench to transmit the required test load between the wrench and the restrained mandrel. This shall be done a total of three times on each mandrel, rotating the wrench 30° to the next gripping position of the wrench after each torque cycle. If the corners of the mandrel are deformed so that the test load values cannot be met, the wrench shall be rejected.

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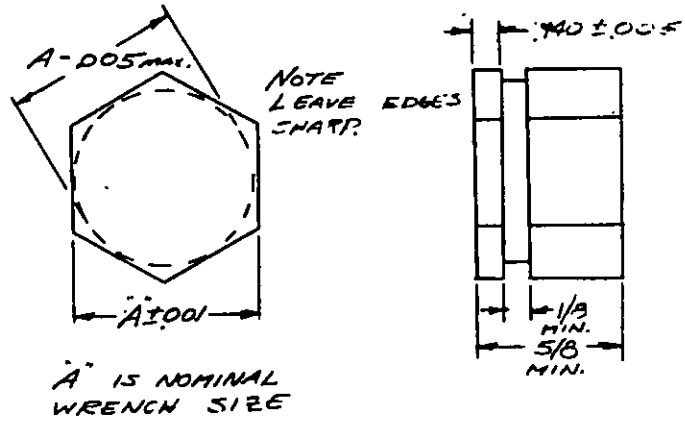


FIGURE 7. Specimen for aluminum mandrel test.

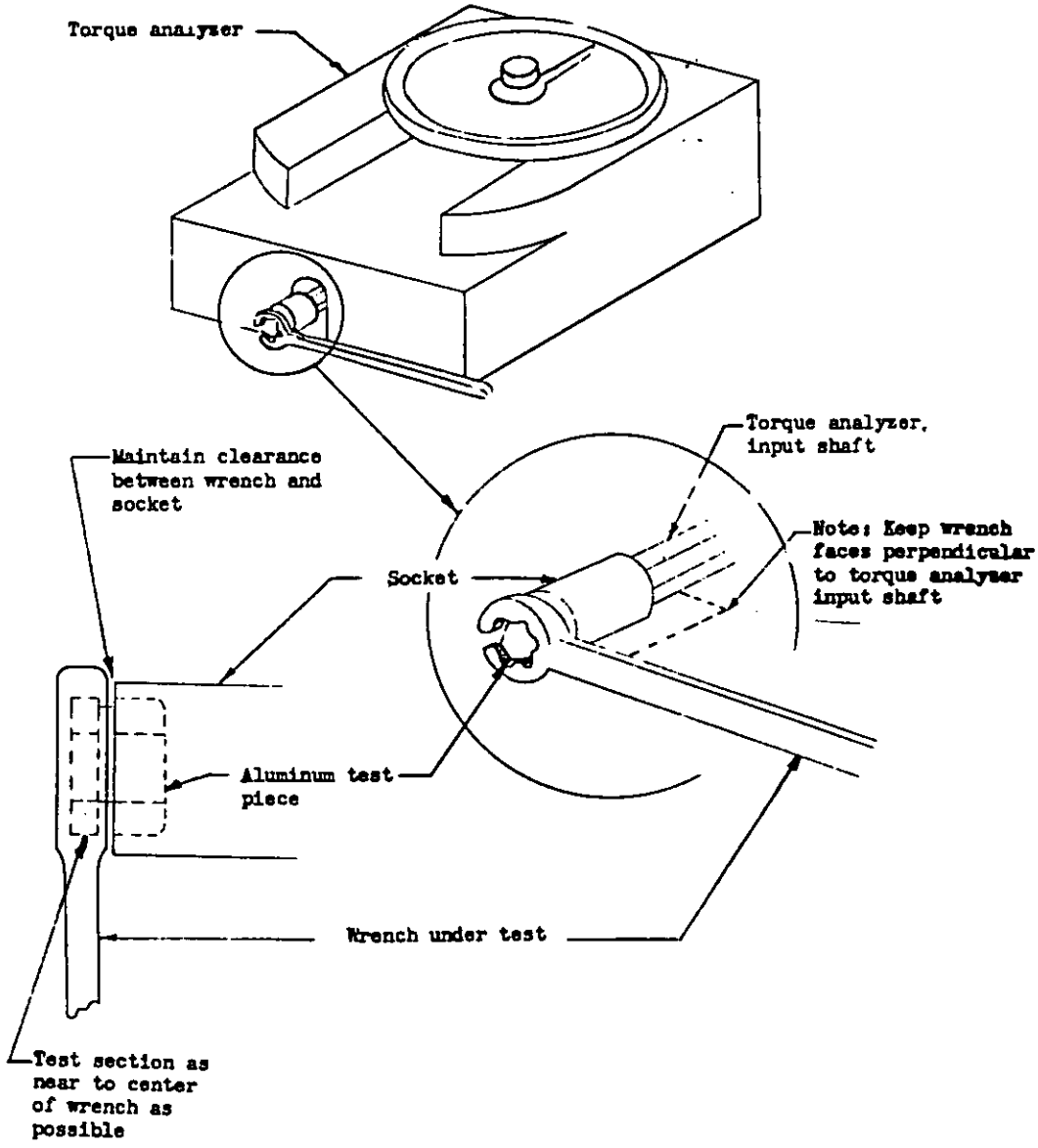


FIGURE 8. Typical test set-up for aluminum mandrel test.

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4.7 Inspection of preparation for delivery. Inspection to determine compliance with preparation for delivery requirements shall be accomplished in accordance with MIL-H-15424.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C, as specified (see 6.1).

5.1.1 Level A. Level A cleaning, preservation, and packaging shall be in accordance with level A of MIL-H-15424.

5.1.2 Level B. Level B preservation and packaging shall be in accordance with level B of MIL-H-15424.

5.1.3 Level C. Preservation and packaging which will afford adequate protection against corrosion, deterioration, and damage during shipment from supply source to the first receiving activity for immediate use (vendor to user).

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Level A packing shall be in accordance with level A of MIL-H-15424.

5.2.2 Level B. Level B packing shall be in accordance with level B of MIL-H-15424.

5.2.3 Level C. Packing which will afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking specified in the contract or order, each unit and intermediate package and shipping container shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options offered herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of wrench required (see 1.2.1).
(c) Size of wrench(s) required (see applicable tables).
(d) If chromeplate is required (see 3.6.2).
(e) Specify if singular sizes or complete sets are required (see 3.7).
(f) If wrench sets are required, specify whether fabric wraps, rolls, or metal boxes are desired (see 3.7).
(g) Selection of applicable levels of preservation, packaging, and packing required (see 5.1 and 5.2).
(h) Additional marking, if required (see 5.3.1).

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